

7 Insurance with Asymmetric Information

This section is based on Chapters 19 from Ljungqvist and Sargent (2012) and Thomas and Worrall (1990).

Environment

- Risk-neutral lender and risk-averse HHs (borrower)
- Lender can borrow or lend outside the economy at β^{-1}
- HH's income y_s is i.i.d. over time and across HHs
- Both the lender and the HH can commit to long-term contracts
- Incentive problem: HH's income is unobservable to the lender $\Rightarrow$ any insurance payments between the lender and the HH must be based on the HHs own reports about their income realizations

An *incentive-compatible* contract makes HHs choose to report their income truthfully.

Revelation principle: if there is any way in which the lender can provide insurance for the HH, there is an equivalent incentive-compatible way in which the HH reports their true income.

Utility Function

- $u : (a, \infty) \rightarrow \mathbb{R}, u \in C^2$
- $u' > 0, u'' < 0$
- $\sup_c u(c) < \infty, \inf_c u(c) = -\infty, \lim_{c \rightarrow a} u'(c) = \infty$
- Thomas and Worrall (1990) also use: $-\frac{u''}{u'}$ is non-increasing (DARA) (sufficient condition to make the value function concave)

Notation

- v : promised value to the HH
- $P(v)$: value to the lender
- b_s : transfer from the lender to the HH if the HH reports $\bar{y}_s$ (repayments to the lender correspond to $b_s < 0$)
- w_s : continuation utility if the HH reports $\bar{y}_s$

First Best

$$P^*(v) = \max_{\{b_s, w_s\}} \sum_s \pi_s [-b_s + \beta P^*(w_s)]$$

$$\text{s.t.} \quad \sum_s \pi_s [u(\bar{y}_s + b_s) + \beta w_s] \geq v$$

$$\Rightarrow \bar{c} : \frac{u(\bar{c})}{1-\beta} = v$$

Second Best (Constrained Optimum)

$$P(v) = \max_{\{b_s, w_s\}} \sum_{s=1}^S \pi_s [-b_s + \beta P(w_s)]$$

$$\text{s.t.} \quad \sum_{s=1}^S \pi_s [u(\bar{y}_s + b_s) + \beta w_s] = v \quad (PK)$$

$$u(\bar{y}_s + b_s) + \beta w_s \geq u(\bar{y}_s + b_k) + \beta w_k, \quad s = 1, \dots, S, \quad k = 1, \dots, S \quad (IC)$$

$$b_s \in (a - \bar{y}_s, \infty), \quad s = 1, \dots, S$$

$$w_s \in (-\infty, v_{max}), \quad s = 1, \dots, S$$

where s = actual state, k = reported state, $v_{max} = \sup_c \frac{u(c)}{1-\beta} < \infty$.

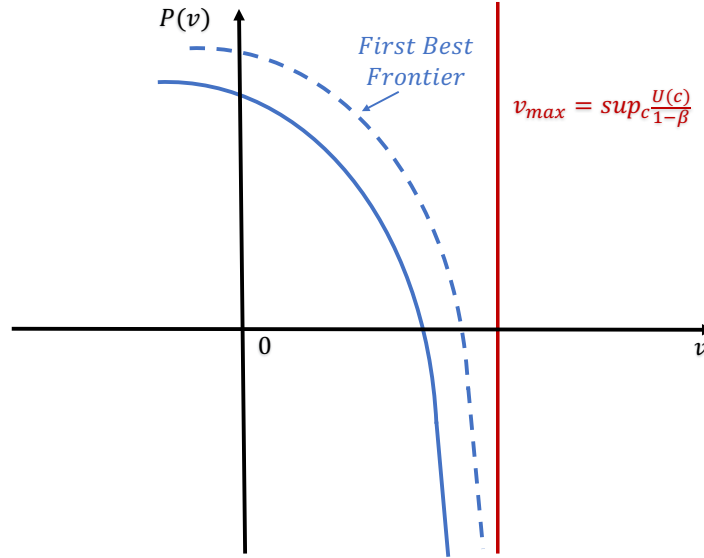
(PK) must be imposed with equality because if the lender awards utility strictly higher than v , it may violate the HH's incentives to report truthfully in earlier periods.

There is no need for a (PC) constraint in this problem because both parties can commit to the contract.

Properties of $P(v)$

- $\lim_{v \rightarrow -\infty} P'(v) = 0$ because $\lim_{c \rightarrow a} u'(c) = \infty$. It becomes very cheap to increase the HH's promised value. u unbounded from below plays a role.
- $\lim_{v \rightarrow v_{max}} P'(v) = -\infty$, and $\lim_{v \rightarrow v_{max}} P(v) = -\infty$ (the opposite true here, since the HH's marginal utility is very low for v close to v_{max})

Figure 1: Properties of $P(v)$



Take two *IC* for local downward and upward deviations. When state is s and $s - 1$, respectively

$$u(\bar{y}_s + b_s) + \beta w_s \geq u(\bar{y}_s + b_{s-1}) + \beta w_{s-1}, \quad (*)$$

$$u(\bar{y}_{s-1} + b_{s-1}) + \beta w_{s-1} \geq u(\bar{y}_{s-1} + b_s) + \beta w_s. \quad (**)$$

Adding them up

$$u(\bar{y}_s + b_s) - u(\bar{y}_{s-1} + b_s) \geq u(\bar{y}_s + b_{s-1}) - u(\bar{y}_{s-1} + b_{s-1})$$

$\Rightarrow b_s \leq b_{s-1}$ by concavity of $u(c)$. Then, from $(*)$ $w_s \geq w_{s-1}$.

That is, a borrower who has (and reports) a lower income receives a higher transfer (provides insurance) from the lender in exchange for a lower continuation utility (provides incentives).

Local upward and downward constraints

Denote $C_{s,k} = u(\bar{y}_s + b_s) + \beta w_s - u(\bar{y}_s + b_k) - \beta w_k$.

It turns out that we do not need all the $C_{s,k} \geq 0$ constraints. In fact, local upward and downward constraints are enough: $C_{s,s+1} \geq 0, C_{s,s-1} \geq 0 \forall s \Rightarrow C_{s,k} \geq 0 \forall s, k$

Proof: Suppose $C_{s,k} \geq 0$ holds for some $s > k$. We will show that this together with $C_{s+1,s} \geq 0$ (local downward) implies $C_{s+1,k} \geq 0$.

We have:

$$u(\bar{y}_s + b_s) + \beta w_s \geq u(\bar{y}_s + b_k) + \beta w_k. \quad (1)$$

Also, as we have shown, $b_s \leq b_k$, hence by concavity of $u(c)$ we have

$$u(\bar{y}_{s+1} + b_s) - u(\bar{y}_s + b_s) \geq u(\bar{y}_{s+1} + b_k) - u(\bar{y}_s + b_k). \quad (2)$$

Moreover, $C_{s+1,s} \geq 0$ means

$$u(\bar{y}_{s+1} + b_{s+1}) + \beta w_{s+1} \geq u(\bar{y}_{s+1} + b_s) + \beta w_s. \quad (3)$$

Summing up (1) and (2), we have

$$u(\bar{y}_{s+1} + b_s) + \beta w_s \geq u(\bar{y}_{s+1} + b_k) + \beta w_k,$$

and using (3),

$$u(\bar{y}_{s+1} + b_{s+1}) + \beta w_{s+1} \geq u(\bar{y}_{s+1} + b_s) + \beta w_s \geq u(\bar{y}_{s+1} + b_k) + \beta w_k$$

$\Rightarrow C_{s+1,k} \geq 0$ holds.

This shows by induction that all global downward constraints are satisfied when the local downward constraints hold.

Analogously, we can show that, starting from any $C_{k,s} \geq 0, k < s$, the local upward constraint $C_{k-1,k} \geq 0$ implies $C_{k-1,s} \geq 0$, and so forth.

This shows that all *IC* can be replaced by $C_{s,s-1} \geq 0$ and $C_{s,s+1} \geq 0 \forall s$.

Only local downward IC bind

Thomas and Worrall (1990) analyze finite horizon $(P_k(v))$ and then take the limit $P(v) = \lim_{k \rightarrow \infty} P_k(v)$. They show that $P_k(v)$ is strictly concave. Hence $P(v)$ is concave.

We will state the following result for the infinite-horizon case, and when proving

it, for simplicity, we will just assume that $P(v)$ is strictly concave (their main result—immiseration—only requires concavity of P).

Claim. *At the optimum, local downward IC bind, and local upward IC never bind. I.e. the HH is always indifferent between reporting his true income and saying that it is a little lower, but never wants to report that it is higher than it actually is.*

Proof: We will only prove that local downward IC bind (see the paper for the proof of the second part). Suppose $C_{k,k-1} > 0$ for some k :

$$u(\bar{y}_k + b_k) + \beta w_k > u(\bar{y}_k + b_{k-1}) + \beta w_{k-1}.$$

Since $b_k \leq b_{k-1} \Rightarrow w_k > w_{k-1}$.

Change $(b_s, w_s)_{s=1}^S$ as follows:

- Keep w_1 fixed
- Reduce w_2 until $C_{2,1} = 0$
- Reduce w_3 until $C_{3,2} = 0$
- $\vdots$

and so on until $C_{s,s-1} = 0 \forall s$

- Add a constant to all w_s to keep $\sum_{s=1}^S \pi_s w_s$ unchanged

The new contract offers the HH the same utility. It is IC because $b_s \leq b_{s-1}$ and $C_{s,s-1} = 0$ together imply that the local upward IC $C_{s-1,s} \geq 0$ is satisfied:

$$u(\bar{y}_s + b_s) + \beta w_s = u(\bar{y}_s + b_{s-1}) + \beta w_{s-1} \Rightarrow$$

$$\begin{aligned} \beta(w_s - w_{s-1}) &= u(\bar{y}_s + b_{s-1}) - u(\bar{y}_s + b_s) \\ &\leq u(\bar{y}_{s-1} + b_{s-1}) - u(\bar{y}_{s-1} + b_s) \end{aligned}$$

where the inequality follows because $b_s \leq b_{s-1}$ and u is concave. At the same time, each $(w_s - w_{s-1})$ has either been reduced or left the same. We generated mean-preserving decrease in continuation values. Since $P(v)$ is strictly concave $\sum_{s=1}^S \pi_s P(w_s) \uparrow \Rightarrow P(v) \uparrow$.

Coinsurance

The optimal contract is characterized by coinsurance: both the HH's and the lender's values increase with a higher income realization

$$(i) \quad u(\bar{y}_s + b_s) + \beta w_s > u(\bar{y}_{s-1} + b_{s-1} + \beta w_{s-1})$$

$$(ii) \quad -b_s + \beta P(w_s) \geq -b_{s-1} + \beta P(w_{s-1})$$

Proof: $\bar{y}_s > \bar{y}_{s-1}$, $c_{s,s-1} = 0 \Rightarrow (i)$. Suppose (ii) is violated, i.e., $-b_s + \beta P(w_s) < -b_{s-1} + \beta P(w_{s-1})$. Then replacing (b_s, w_s) in the contract by (b_{s-1}, w_{s-1}) raises the lender's profits but leaves the HH's utility unchanged because $c_{s,s-1} = 0$, and the change is also *IC*. Thus, the optimal contract is such that the lender's profits weakly increase in the HH's income realization.

We can now write the lender's problem as follows:

$$\begin{aligned} P(v) &= \max_{\{b_s, w_s\}} \sum_{s=1}^S \pi_s [-b_s + \beta P(w_s)] \\ \text{s.t.} \quad &\sum_{s=1}^S \pi_s [u(\bar{y}_s + b_s) + \beta w_s] = v \quad (\lambda) \\ &u(\bar{y}_s + b_s) + \beta w_s \geq u(\bar{y}_s + b_{s-1}) + \beta w_{s-1}, \quad s = 2, \dots, S. \quad (\mu_s) \end{aligned}$$

F.O.C.

$$[b_s] : \pi_s [1 - \lambda u'(\bar{y}_s + b_s)] = \mu_s u'(\bar{y}_s + b_s) - \mu_{s+1} u'(\bar{y}_{s+1} + b_s) \quad (4)$$

$$[w_s] : \pi_s [P'(w_s) + \lambda] = \mu_{s+1} - \mu_s \quad (5)$$

for $s = 1, 2, \dots, S$, where $\mu_1 = \mu_{S+1} = 0$

$$[\text{Env}] : P'(v) = -\lambda \quad (6)$$

Summing up (5) over s and using $\sum_{s=1}^S \mu_{s+1} - \mu_s = \mu_{S+1} - \mu_1 = 0$ and (5), yields

$$\sum_{s=1}^S \pi_s P'(w_s) = P'(v). \quad (7)$$

That is, $P'(v_t)$ is a martingale (we can think of $\{v_t\}$, the sequence of the HH's promised values, as a stochastic process which depends on realizations of income over time).

Note: A (discrete-time) **martingale** is a sequence $X_1, X_2, X_3, \dots$ of integrable random variables satisfying $E[X_{t+1}|X_1, \dots, X_t] = X_t$; a **submartingale** if $E[X_{t+1}|X_1, \dots, X_t] \geq X_t$; a **supermartingale** if $E[X_{t+1}|X_1, \dots, X_t] \leq X_t$.

Spreading continuation values

Claim. $w_1 < v < w_S$

Proof: Suppose that $w_S \leq v$. Since $w_s \geq w_S \forall s$ and $P(v)$ is concave, (7) implies that $P'(w_s) = P'(v) \forall s$.

Then, substituting (6) into (5): $\pi_s[P'(w_s) - P'(v)] = \mu_{s+1} - \mu_s$, for $s = 2$, $\mu_2 = 0$ and for $s = S$, $-\mu_S = 0$.

Using this in (5), we conclude that $\mu_s = 0 \forall s \Rightarrow$ using (4), this implies that $u'(\bar{y}_s + b_s) = \frac{1}{\lambda} \forall s \Rightarrow b_{s-1} > b_s$, as $\bar{y}_{s-1} < \bar{y}_s$.

But at the same time, $w_{s-1} = w_s$ and IC, $u(\bar{y}_s + b_s) + \beta w_s \geq u(\bar{y}_s + b_{s-1}) + \beta w_{s-1} \Rightarrow b_s \geq b_{s-1}$, which is a contradiction. Thus, $w_S > v$.

A symmetric argument shows that $w_1 < v$.

That is, the lender must spread out future promised utilities in order to provide both insurance today and incentives not to misreport. How the lender balances w_s 's compared to v is given by (7).

Immiseration Result

Expression (7) also contains the key to the long-run behavior of v_t .

Claim. $\lim_{t \rightarrow \infty} v_t = -\infty$ *almost surely*.

This is often referred as the immiseration result.

Sketch of the proof:

- $P'(v)$ is a (non-positive) martingale
- **Doob's martingale convergence theorem:** Let $\{X_n\}$ be a submartingale such that $\sup_n E[X_n^+] < \infty$, where X_n^+ is the positive part of X_n , i.e., $X_n^+ = \max\{X_n, 0\}$. Then as $n \rightarrow \infty$, X_n converges almost surely to a random variable X with finite expectation.¹

Note that with a *non-positive* (sub)martingale, $\sup_n E[X_n^+] = 0 < \infty$.

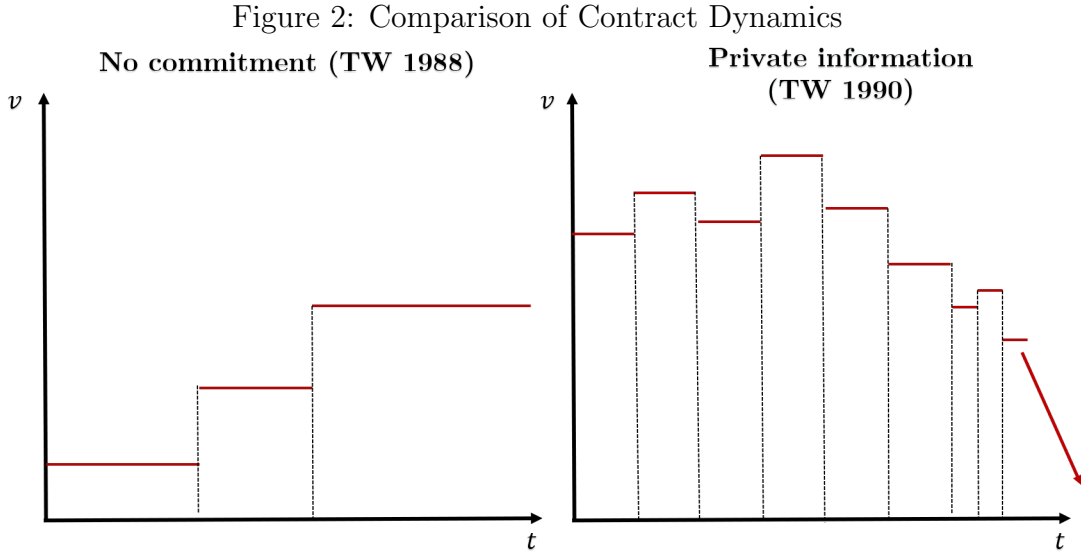
¹There is a symmetric statement for supermartingales with bounded expectation of the positive part: Let $\{X_n\}$ be a supermartingale such that $\sup_n E[X_n^-] < \infty$, where X_n^- is the negative part of X_n , i.e., $X_n^- = -\min(X_n, 0)$. Then as $n \rightarrow \infty$, X_n converges almost surely to a random variable X with finite expectation.

- By the martingale convergence theorem, $P'(v) \rightarrow X$ a.s.
- The rest of the proof shows that $X = 0$ a.s.
- Thus, $P'(v) \rightarrow 0$ a.s. $\Rightarrow v_t \rightarrow -\infty$ a.s.

Intuition

The idea behind showing that $X = 0$ is as follows. Since future promised values have to be spread out, $\{v_t\}$ cannot converge to a finite limit. The only way of not getting a contradiction is to have $v_t \rightarrow -\infty$.

We also have that $\lim_{v \rightarrow v_{max}} P'(v) = -\infty$, $\lim_{v \rightarrow -\infty} P'(v) = 0$. The cost of spreading w_1 and w_S is lower if w_1 and w_S are themselves lower (*IC* is cheaper if promised values are lower). A contract in which v declines over time can induce the HH to tell the truth by using large variability in future utility and at the same time smooth consumption in the initial periods.



As we see in Figure 2 in contracts under no commitment v_t never decreases, while under private information v_t can decrease or decrease and have a downward drift.

Atkeson and Lucas (1992)

- General Equilibrium (closed economy), where the constrained efficient allocation also has HH's expected utility converging to the minimum level with probability 1.

- Planner chooses and IC allocation subject to a constraint that the total consumption $\leq$ endowment in each period
- Unobserved idiosyncratic taste shocks ϵ (i.i.d. across time and HHs).
- Per period utility: $u(c, \epsilon) = \epsilon \log(c)$
 - or $u(c, \epsilon) = \epsilon \frac{c^\gamma}{\gamma}$, $\gamma < 1, \gamma \neq 0$
 - or $u(c, \epsilon) = -\epsilon e^{-\gamma c}$, $\gamma > 0$

They show that an efficient allocation is such that a decreasing over time fraction of the population is receiving an increasing over time fraction of the total resources. I.e., inequality is growing over time; in the limit, mass 1 is in misery, and mass 0 are infinitely rich.

Cole and Kocherlakota (2001)

They extended the model to allow for unobservable savings on the part of the agent (storage) with return R not necessarily equal to the discount factor (but they need $R \geq \beta^{-1}$).

The model now has adverse selection (unobservable income) and moral hazard (unobservable storage) and it cannot be easily defined recursively.

The main results are as follows:

- The optimal contract delivers the same net present value of transfer (income minus consumption) to all types, independent of their histories.
 - The main trade-off: $IC \Rightarrow$ You need to give more NPV to agents with low incomes. But effective risk sharing demands the opposite. At the optimum, both forces cancel out and NPV is constant
- The symmetric effective allocation of consumption is decentralizable through a competitive asset market in which individuals trade risk-free bonds among themselves.

References

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